Three dimension coordinate system (12.1)

Aug-25-2025

3d Coodinate system

equation review

Pythagoreon theorm:

$$a^2 + b^2 = c^2$$

distance Formula

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

midPoint Formula:

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

equation of a Circle:

$$(x - x_1)^2 + (y - y_1)^2 = r^2$$

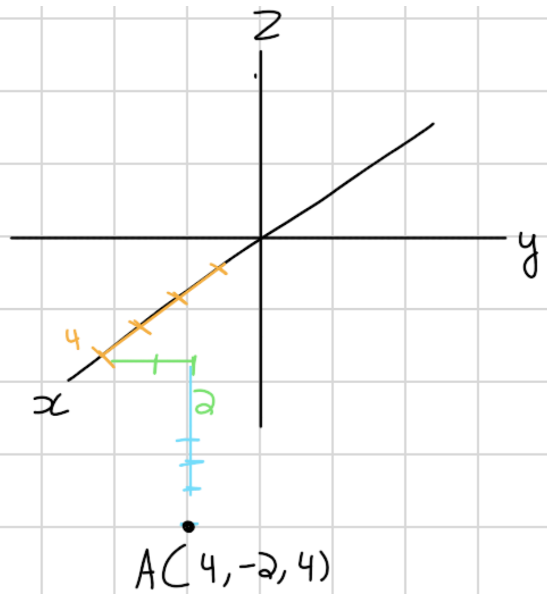
equations and Formula's

Formulas	2d equations	3d equations
Distance between two Points	$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$	$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$
equation of a circle/sPhere	$(x-x_1)^2+(y-y_1)^2=r^2$	$(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=r^2$
midPoint	$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$	$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$

Made with Goodnotes

graph

A(4,2,-4)



- began by Plotting "x" then go to to "y" and end with "z"

★ You need to use Completing the Square

Quick review on Completing the square

$x^2 + 6x - 7 = 0$ • Move the "7" to the other Side

$x^2 + 6x \_\_ = 7$ • take "6" and divide it by 2 and Square the answer and Place it in "__" and what every you do on one Side you must do on the other

$x^2 + 6x + 9 = 7 + 9$

$x^2 + 6x + 9 = 16$

$(x+3)^2 = 16$ • the three Came From taking the middle CoefFcient and dividing it by 2

Practice Problem #1

$(x-1)^2 + y^2 + 6y^2 + z^2 - 5z = 9$

$(x-1)^2 + y^2 + 6y + \square + z^2 - 5z + \square = 9$

apply Completing the square

$(x-1)^2 + y^2 + 6y + \boxed{9} + z^2 - 5z + \boxed{\frac{25}{2}} = 9 + 9 + \frac{25}{5}$

$(x-1)^2 + y^2 + 6y + \boxed{9} + z^2 - 5z + \boxed{\frac{25}{2}} = \frac{97}{4}$ • this in the Sphere equation was Square so to get the radius you need to apply Square root

Center $= (1, -3, \frac{5}{2})$ $r = \sqrt{\frac{97}{4}}$

Sphere equation

$$(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=r^2$$

So if you what to get the center but in on of the Paraenthesis is a addition sign that number is a negitive

• Find the distance between $D(4,-2,-1)$ and yz- Plane

to solve this you can do this in a simple way you are given $D(4,-2,-1)$ and the question is is asking the distance between the Point and the yz-Plane you have letters yz but no "x" the answer is the missing letter So its 4

✰ this also work on any question given to you just find the missing letter

• Find the distance between $D(4,-2,-1)$ and y-axis

$D(4,-2,-1)$ $(0,-2,0)$

• given equation • y-axis

✰ use distance equation

$$\text{distance} = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

$\underset{x_1\ y_1\ z_1}{(4,-2,-1)}$ $\underset{x_2\ y_2\ z_2}{(0,-2,0)}$

$$\sqrt{(0-4)^2+(2-(-2))^2+(0-(-1))^2} = \boxed{\sqrt{33}}$$

✰ If you are given a question asking about the x-axis then $(4,0,0)$ and do the same exact thing with distance formula if another question asks about z-axis $(0,0,-1)$ and do the same thing with the distance formula

• given a sphere has a center $C(1,4,3)$ and has a Point $B(0,2,-3)$ Find the equation of the equation of sphere

Find the radius using distance formula

$$\text{distance} = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

$\underset{x_1\ y_1\ z_1}{(1,4,3)}$ $\underset{x_2\ y_2\ z_2}{(0,2,-3)}$

$$\sqrt{(0-1)^2+(2-4)^2+(-3-3)^2} = \sqrt{41}$$

✰ the center is what we use in the sphere equation

$$(x-x_1)^2+(y-y_1)^2+(z-z_1)^2 = r^2$$

Center = $(1,4,3)$

$$\boxed{(x-1)^2+(y-4)^2+(z-3)^2 = 41}$$

Find the equation of the sphere with diameter that has the end points $(5,-4,-3)$ and $(3,-1,2)$

began by Finding the center using the midpoint formula

$$\left(\frac{x_2+x_1}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

$\underset{x_1\ y_1\ z_1}{(5,-4,-3)}$ $\underset{x_2\ y_2\ z_2}{(3,-1,2)}$

Center = $(4, -\frac{5}{2}, -\frac{1}{2})$

Find the radius using the distance formula

$$\text{distance} = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

$\underset{x_1\ y_1\ z_1}{(5,-4,-3)}$ $\underset{x_2\ y_2\ z_2}{(3,-1,2)}$

$$\sqrt{(3-5)^2+(-1-(-4))^2+(2-(-3))^2}$$

$= \sqrt{38}$

the question said it gave you diameter so you need to divide by two to get radius

radius = $\frac{\sqrt{32}}{2}$

You need to use the center you got from the midpoint formula to get center to plug into the sphere formula

Center = $(4, -\frac{5}{2}, -\frac{1}{2})$

Sphere equation:

$$(x-x_1)^2+(y-y_1)^2+(z-z_1)^2 = r^2$$

$$\boxed{(x-4)^2+(y+\tfrac{5}{2})^2+(z+\tfrac{1}{2}) = 8}$$